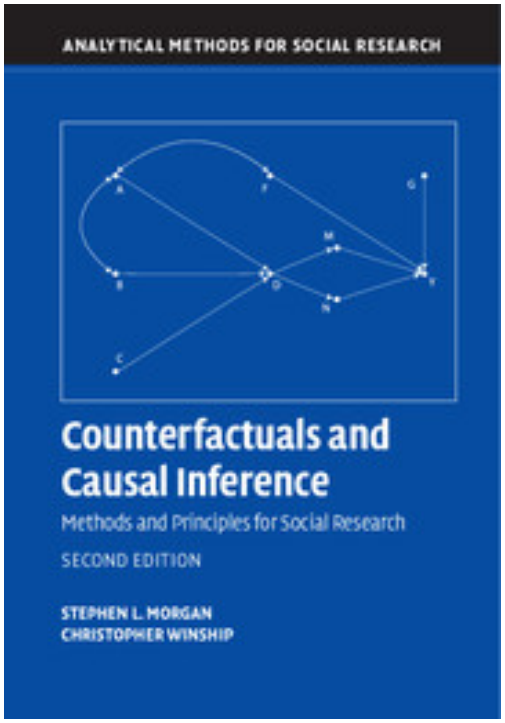
Counterfactuals and Causal Inference

Methods and Principles for Social Research



Chapter 9

Outline

1. Causal Effect Estimation with a Binary IV
2. Traditional IV Estimators
3. Instrumental Variable Estimators in the Presence of Individual-Level Heterogeneity
4. Conclusions

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The traditional IV estimator (or the Wald estimator)

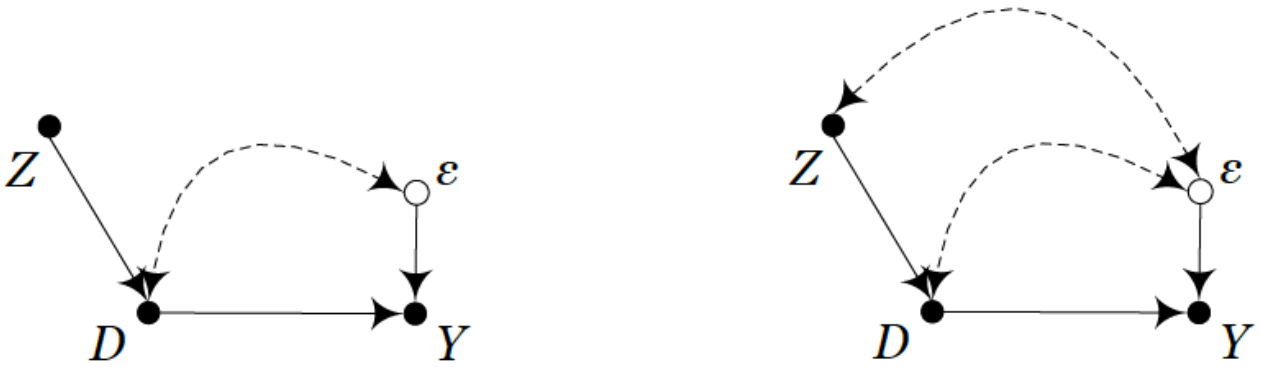
- The simplest scenario in which an instrumental variable can be used to effectively estimate a causal effect

$$Y = \alpha + \delta D + \varepsilon, \tag{9.1}$$

  where Y is the outcome variable, D is a binary causal exposure variable, α is an intercept, δ is the causal effect of D on Y, and ε is a summary random variable that represents all other causes of Y.
  - In Pearl's framework, we are assuming that $f_Y(D, e_Y)$ is defined as the linear function $\alpha + \delta D + \varepsilon$ and where α and β are scalar constants.
- Suppose that D and ε are correlated, as in the regression chapter, so that bivariate regression will not yield a consistent or unbiased estimate of δ.

The heart of the matter for traditional IV estimation

- If we maintain the assumption that the effect of D on Y is a constant structural effect δ, then it is not necessary to relate all of the variation in D to all of the variation in Y in order to obtain a consistent estimate of the causal effect.
- If we can find a way of isolating the covariation in D and Y that is causal, then we can ignore the other covariation in D and Y that is noncausal because it is generated by the common causes of D and ε.
- An "instrument" for D that selects out this causal covariation is what we need.



(a) Z is a valid instrumental variable for D

(b) Z is not a valid instrumental variable for D

Figure 9.1 Two graphs in which Z is a potential instrumental variable.

To see this result without reference to graphs, take the population-level expectation of Equation (9.1), $E[Y] = E[\alpha + \delta D + \varepsilon] = \alpha + \delta E[D] + E[\varepsilon]$, and rewrite it as a difference equation in Z:

$$\begin{aligned} & E[Y|Z=1] - E[Y|Z=0] \\ & = \delta(E[D|Z=1] - E[D|Z=0]) + (E[\varepsilon|Z=1] - E[\varepsilon|Z=0]) \end{aligned} \tag{9.2}$$

Take Equation (9.2) and divide both sides by $E[D|Z=1] - E[D|Z=0]$, yielding

$$\begin{aligned} &\frac{E[Y|Z=1] - E[Y|Z=0]}{E[D|Z=1] - E[D|Z=0]} \\ &= \frac{\delta(E[D|Z=1] - E[D|Z=0]) + (E[\varepsilon|Z=1] - E[\varepsilon|Z=0])}{E[D|Z=1] - E[D|Z=0]} \end{aligned} \tag{9.3}$$

If the data are generated by the set of causal relationships depicted in Figure 9.1(a), then Z has no linear association with ε, and $E[\varepsilon|Z=1]-E[\varepsilon|Z=0]$ in Equation (9.3) is equal to 0.

Consequently, the right-hand side of Equation (9.3) simplifies to δ:

$$\frac{E[Y|Z=1]-E[Y|Z=0]}{E[D|Z=1]-E[D|Z=0]}=\delta \tag{9.4}$$

If Z is in fact associated with D but not associated with ε (or with Y, except through D), then the following sample-based estimator will equal δ in an infinite sample:

$$\hat{\delta}_{\text{IV,WALD}} \equiv \frac{E_N[y_i|z_i = 1] - E_N[y_i|z_i = 0]}{E_N[d_i|z_i = 1] - E_N[d_i|z_i = 0]} \tag{9.5}$$

For completeness, return to consideration of Figure 9.1(b), in which Z has a nonzero association with ε.

In this case, $E[\varepsilon|Z=1]-E[\varepsilon|Z=0]$ in Equations (9.2) and (9.3) cannot be equal to 0, and thus Equation (9.3) does not reduce further to Equation (9.4). Rather, it reduces only to

$$\begin{aligned}\frac{E[Y|Z=1]-E[Y|Z=0]}{E[D|Z=1]-E[D|Z=0]} \\ = \delta + \frac{E[\varepsilon|Z=1]-E[\varepsilon|Z=0]}{E[D|Z=1]-E[D|Z=0]}\end{aligned} \tag{9.6}$$

IV Demonstration 1

- A random sample of 10,000 ninth graders, $\{y_i, d_i\}_{i=1}^{10,000}$, where Y is the score on the standardized test and D is equal to 1 for students who attend private high schools and equal to 0 for students who attend public high schools.
- Suppose that the values of y_i are regressed on the values of d_i and that a predicted regression surface is obtained:

$$\hat{Y} = 50.0 + 9.67\,(D)\,. \tag{9.7}$$

The voucher IV

- All eighth graders in the city (both in public and private schools) are entered into a random lottery for \$3,000 school vouchers that are redeemable at a private high school. By mandate, 10 percent of all eligible students win the voucher lottery.
- After merging the data, the state officials cross-tabulate eighth grade lottery winners (where $z_i = 1$ for those who won the lottery and $z_i = 0$ for those who did not) by school sector attendance in the ninth grade, d_i.

Table 9.1 The Distribution of Voucher Winners by School Sector for IV Demonstration 1

		Public school $d_i = 0$	Private school $d_i = 1$
Voucher loser	$z_i = 0$	8000	1000
Voucher winner	$z_i = 1$	800	200

- The dummy variable Z for winning the voucher lottery is a valid IV for D because
  1. the randomization of the lottery renders Z independent of ε in the population-level causal regression equation $Y = \alpha + \delta D + \varepsilon$
  2. Z has a causal effect on Y only through D
- The $\hat{\delta}_{\text{IV,WALD}}$ is

$$\frac{E_N[y_i|z_i = 1] - E_N[y_i|z_i = 0]}{E_N[d_i|z_i = 1] - E_N[d_i|z_i = 0]} = \frac{51.600 - 51.111}{.200 - .111} = 5.5 \quad (9.8)$$

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The linear structural equation IV estimator

- Recall the causal regression setup in Equation (9.1):

$$Y = \alpha + \delta D + \varepsilon, \tag{9.9}$$

  and again assume that we are in the traditional setup where δ is an invariant, structural causal effect that applies to all members of the population of interest.
- The ordinary least squares (OLS) estimator of the regression coefficient on D is

$$\hat{\delta}_{\text{OLS, bivariate}} \equiv \frac{\text{Cov}_N(y_i, d_i)}{\text{Var}_N(d_i)} \tag{9.10}$$

- Suppose that the back-door association between D and ε renders the least squares estimator biased and inconsistent for δ in Equation (9.9)
- If least squares cannot be used to effectively estimate δ, an alternative IV estimator can be attempted, with an IV Z, as in

$$\hat{\delta}_{\mathrm{IV}} \equiv \frac{\mathrm{Cov}_N(y_i, z_i)}{\mathrm{Cov}_N(d_i, z_i)}, \tag{9.11}$$

- The covariance between the outcome Y and the instrument Z can be written as

$$\mathrm{Cov}(Y, Z) = \delta\,\mathrm{Cov}(D, Z) + \mathrm{Cov}(\varepsilon, Z) \tag{9.12}$$

  again assuming that δ is a constant structural effect.
- Dividing by $\mathrm{Cov}(D, Z)$ then yields

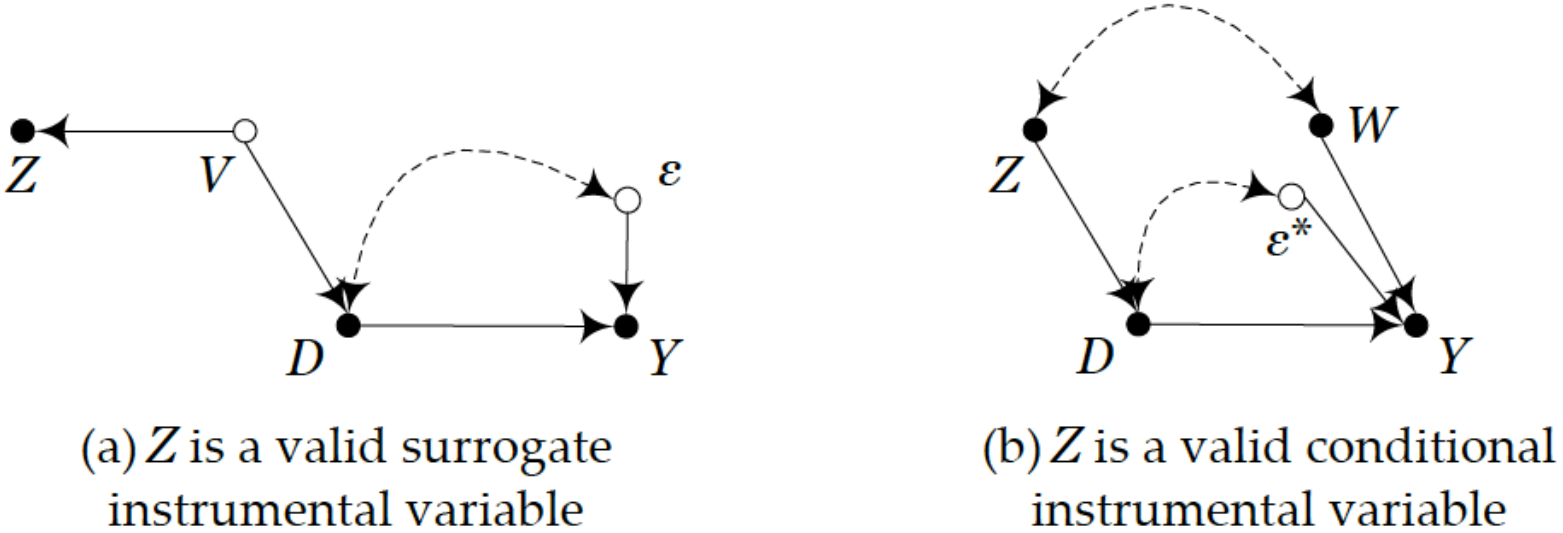
$$\frac{\mathrm{Cov}(Y, Z)}{\mathrm{Cov}(D, Z)} = \frac{\delta\,\mathrm{Cov}(D, Z) + \mathrm{Cov}(\varepsilon, Z)}{\mathrm{Cov}(D, Z)} \tag{9.13}$$

- When $\mathrm{Cov}(\varepsilon, Z)$ is equal to 0 in the population, then the right-hand side of Equation (9.13) simplifies to δ. This suggests that

$$\frac{\mathrm{Cov}_N(y_i, z_i)}{\mathrm{Cov}_N(d_i, z_i)} \xrightarrow{p} \delta \tag{9.14}$$

  if $\mathrm{Cov}(\varepsilon, Z) = 0$ in the population and if $\mathrm{Cov}(D, Z) \neq 0$.
- This would be the case for the causal diagram in Figure 9.1(a). But here the claim is more general and holds for cases in which Z is many-valued (and, in fact, for cases in which D is many-valued as well, assuming that the linear, constant-coefficient specification in Equation (9.9) is appropriate).

Surrogate and conditional IVs



(a) Z is a valid surrogate instrumental variable

(b) Z is a valid conditional instrumental variable

Figure 9.2 Two graphs in which Z is a valid IV.

Some examples of IV estimation

- For the effect of education on earnings: proximity to college, regional and temporal variation in school construction, tuition at local colleges, temporal variation in the minimum school-leaving age, and quarter of birth
- Military service on earnings: Vietnam era draft lottery

The most crucial IV identifying assumption cannot be tested

- The basic identification assumption underlying all of the studies just summarized – that Z has no net association with Y except for the one generated by the directed path $Z \rightarrow D \rightarrow Y$ – is a strong and untestable assumption.
- Some researchers believe mistakenly that this assumption is empirically testable. In particular, they believe that the assumption that Z has no direct effect on Y implies that there is no association between Z and Y conditional on D. Thus, if an association between Z and Y is detected after conditioning on D, then the assumption must be incorrect.

But ... D is a collider

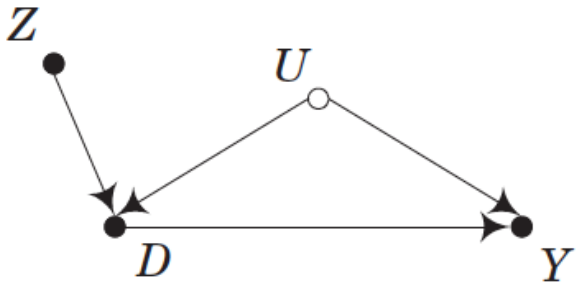


Figure 9.3 A graph with an unblocked back-door path and a valid IV.

Recognized pitfalls of traditional IV estimation

1. The assumption that an IV does not have a net direct effect on the outcome variable is often hard to defend.
2. Even when an IV does not have a net direct effect on the outcome variable, IV estimators are biased in finite samples. Moreover, this bias can be substantial when an instrument only weakly predicts the causal variable.
   - The expected bias in the Wald estimator for a binary IV as the term
   $$\frac{E[\varepsilon|Z=1]-E[\varepsilon|Z=0]}{E[D|Z=1]-E[D|Z=0]} \tag{9.15}$$
   - If the denominator is close to zero, even a tiny violation of the identifying assumption can generate a large amount of bias. And, unfortunately, this relationship is independent of the sample size.

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Embrace the heterogeneity ... and insight will emerge.

- Recall the definition of Y presented at various points as

$$\begin{aligned} Y &= Y^0 + (Y^1 - Y^0)D \\ &= Y^0 + \delta D \\ &= \mu^0 + \delta D + \upsilon^0, \end{aligned} \tag{9.16}$$

where $\mu^0 \equiv E[Y^0]$ and $\upsilon^0 \equiv Y^0 - E[Y^0]$.

  - Note that δ is now defined as $Y^1 - Y^0$, unlike its structural representation in Equations (9.1) and (9.9) where δ was implicitly assumed to be constant across all individuals.

A taxonomy for response to the instrument

When D and Z are binary variables, there are four possible groups of individuals in the population. These can be summarized by a four-category latent variable C for compliance status:

$$\begin{aligned}
\text{Compliers } (C = c): &\quad D^{Z=0} = 0 \text{ and } D^{Z=1} = 1, \\
\text{Defiers } (C = d): &\quad D^{Z=0} = 1 \text{ and } D^{Z=1} = 0, \\
\text{Always takers } (C = a): &\quad D^{Z=0} = 1 \text{ and } D^{Z=1} = 1, \\
\text{Never takers } (C = n): &\quad D^{Z=0} = 0 \text{ and } D^{Z=1} = 0.
\end{aligned}$$

The observed treatment indicator variable D can then be defined as

$$\begin{aligned} D &= D^{Z=0} + (D^{Z=1} - D^{Z=0})Z \\ &= D^{Z=0} + \kappa Z, \end{aligned} \tag{9.17}$$

where $\kappa \equiv D^{Z=1} - D^{Z=0}$.

- The parameter κ in Equation (9.17) is the individual-level causal effect of the instrument on D, and it varies across individuals if $D^{Z=1} - D^{Z=0}$ varies across individuals (i.e., if observed treatment status varies as the instrument is switched from "off" to "on" for each individual).
- If the instrument represents encouragement to take the treatment, such as the randomly assigned school voucher in IV Demonstration 1, then κ can be interpreted as the individual-level compliance inducement effect of the instrument.
  - $\kappa = 1$ for compliers
  - $\kappa = -1$ for defiers
  - $\kappa = 0$ for always takers and never takers

LATE identifying assumptions

Given these definitions of potential outcome variables and potential treatment variables, a valid instrument Z for the causal effect of D on Y must satisfy three assumptions in order to identify a LATE:

$$\text{Independence assumption: } (Y^1, Y^0, D^{Z=1}, D^{Z=0}) \perp\!\!\!\perp Z, \tag{9.18}$$

$$\text{Nonzero effect of instrument assumption: } \kappa \neq 0 \text{ for all } i, \tag{9.19}$$

$$\text{Monotonicity assumption: either } \kappa \geq 0 \text{ for all } i \text{ or } \kappa \leq 0 \text{ for all } i. \tag{9.20}$$

With monotonicity, we have a new result:

- If $\kappa \geq 0$ for all i, then the Wald estimator converges to a particular LATE:

$$\hat{\delta}_{\text{IV,WALD}} \xrightarrow{p} E[\delta|C = c], \tag{9.21}$$

  which is equal to $E[Y^1 - Y^0|D^{Z=1} = 1, D^{Z=0} = 0]$ and is therefore the average treatment effect among compliers.
- In contrast, if $\kappa \leq 0$ for all i, then the Wald estimator converges to the opposite LATE:

$$\hat{\delta}_{\text{IV,WALD}} \xrightarrow{p} E[\delta|C = d], \tag{9.22}$$

  which is equal to $E[Y^1 - Y^0|D^{Z=1} = 0, D^{Z=0} = 1]$ and is therefore the average treatment effect among defiers.

IV Demonstration 2

Table 9.2 The Joint Probability Distribution and Conditional Expectations of the Test Score for Voucher Winner by School Sector for IV Demonstrations 1 and 2

		Public school $d_i = 0$	Private school $d_i = 1$
Voucher loser	$z_i = 0$	$N = 8000$	$N = 1000$
		$\Pr_N[.,.] = .8$	$\Pr_N[.,.] = .1$
		$E_N[y_i \vert .,.] = 50$	$E_N[y_i \vert .,.] = 60$
Voucher winner	$z_i = 1$	$N = 800$	$N = 200$
		$\Pr_N[.,.] = .08$	$\Pr_N[.,.] = .02$
		$E_N[y_i \vert .,.] = 50$	$E_N[y_i \vert .,.] = 58$

Consider how the least squares and IV estimates are calculated

- The coefficient of 9.67 on D in Equation (9.7) is equal to the naive estimator, $E_N[y_i|d_i = 1] - E_N[y_i|d_i = 0]$
- These two conditional expectations can be calculated from the columns of Table 9.2:

$$\begin{aligned} E_N[y_i|d_i = 1] &= \frac{.1}{.1 + .02}60 + \frac{.02}{.1 + .02}58 = 59.667, \\ E_N[y_i|d_i = 0] &= \frac{.8}{.8 + .08}50 + \frac{.08}{.8 + .08}50 = 50.0. \end{aligned}$$

- The IV estimate of 5.5 is the ratio of two specific contrasts:

$$\frac{E_N[y_i|z_i = 1] - E_N[y_i|z_i = 0]}{E_N[d_i|z_i = 1] - E_N[d_i|z_i = 0]} = \frac{51.6 - 51.111}{.2 - .111} = 5.5. \quad (9.23)$$

- Both of these contrasts are calculated within the rows of Table 9.2 rather than the columns.

- The contrast in the numerator is the naive estimate of the effect of Z on Y. It is calculated as the difference between

$$E_N[y_i|z_i = 1] = \frac{.08}{.08 + .02}50 + \frac{.02}{.08 + .02}58 = 51.6 \quad \text{and}$$
$$E_N[y_i|z_i = 0] = \frac{.8}{.8 + .1}50 + \frac{.1}{.8 + .1}60 = 51.111.$$

- The contrast in the denominator is the naive estimate of the effect of Z on D. It is calculated as the difference between

$$E_N[d_i = 1|z_i = 1] = \frac{.02}{.08 + .02} = .2 \quad \text{and}$$
$$E_N[d_i = 1|z_i = 0] = \frac{.1}{.8 + .1} = .111.$$

- Thus, calculating the IV estimate in this case is quite simple and does not require any consideration of the underlying potential outcomes or potential treatments.

But, let's go deeper ...

- The proportion of always takers can be estimated consistently from the first row of Table 9.2, and the proportion of never takers can be estimated consistently from the second row of Table 9.2 as

$$\frac{\Pr_N[d_i = 1, z_i = 0]}{\Pr_N[z_i = 0]} \xrightarrow{p} \Pr[C = a], \tag{9.24}$$

$$\frac{\Pr_N[d_i = 0, z_i = 1]}{\Pr_N[z_i = 1]} \xrightarrow{p} \Pr[C = n] \ . \tag{9.25}$$

- For this example, the proportion of always takers is $1,000/9,000 = .111$, and the proportion of never takers is $800/1,000 = .8$.

- Because no defiers exist in the population with regard to this instrument, these two estimated proportions can be subtracted from 1 in order to obtain the proportion of compliers:

$$1 - \frac{\Pr_N[d_i = 1, z_i = 0]}{\Pr_N[z_i = 0]} - \frac{\Pr_N[d_i = 0, z_i = 1]}{\Pr_N[z_i = 1]} \xrightarrow{p} \Pr[C = c] \ . \tag{9.26}$$

- For this example, the proportion of compliers in the population is $1 - .111 - .8 = .089$.

With a bit of rounding, we can declare ...

Table 9.3 The Distribution of Never Takers, Compliers, and Always Takers for IV Demonstration 2

		Public school $d_i = 0$	Private school $d_i = 1$
Voucher loser	$z_i = 0$	7200 Never takers 800 Compliers	1000 Always takers
Voucher winner	$z_i = 1$	800 Never takers	111 Always takers 89 Compliers

We can write these expressions for the groups on the diagonal of Table 9.3

$$E[Y|D=1,Z=1] = \frac{\Pr[C=c]}{\Pr[C=c]+\Pr[C=a]}E[Y^1|C=c] \quad (9.27)$$
$$+ \quad \frac{\Pr[C=a]}{\Pr[C=c]+\Pr[C=a]}E[Y^1|C=a],$$

$$E[Y|D=0,Z=0] = \frac{\Pr[C=c]}{\Pr[C=c]+\Pr[C=n]}E[Y^0|C=c] \quad (9.28)$$
$$+ \quad \frac{\Pr[C=n]}{\Pr[C=c]+\Pr[C=n]}E[Y^0|C=n].$$

Note: We could write these even in the presence of defiers. Defiers would fall into the off-diaognal.

Plugging in what we can get from the data yields

$$58 = \frac{.089}{.089 + .111} E[Y^1|C = c] + \frac{.111}{.089 + .111} 60 \tag{9.29}$$

$$50 = \frac{.089}{.089 + .8} E[Y^0|C = c] + \frac{.8}{.089 + .8} 50 \tag{9.30}$$

- Solving Equation (9.29) for $E[Y^1|C = c]$ results in 55.5, whereas solving Equation (9.30) for $E[Y^0|C = c]$ results in 50.
- The difference between these values is 5.5, which is the average causal effect for the subset of all students who would attend a private school if given a voucher but would not attend a private school in the absence of a voucher.

The value of 5.5 yields no information whatsoever about the effect of private schooling for the always takers and the never takers.

Implications of the LATE perspective for traditional IV estimation

- If causal effect heterogeneity is present, then a constant-coefficient interpretation of an IV estimate is implausible.
- If the instrument satisfies a monotonicity condition and can be conceptualized as a proximate inducement to take the treatment, then IV estimates can be given LATE interpretations.
- Two-stage least squares is unlikely to resolve complications due to heterogeneity.

2SLS

- One traditional way to bolster a homogeneity assumption is to condition on variables in a vector X that can account for all such heterogeneity and then assert a conditional homogeneity of response assumption.
- The endogenous regressors D and X are embedded in an encompassing $\mathbf{X}$ matrix, which is $n \times k$, where n is the number of respondents and k is the number of variables in X plus 2 (one for the constant and one for the treatment variable D).
- A matrix $\mathbf{Z}$ is constructed that is equivalent to X, except that the column in $\mathbf{X}$ that includes the treatment variable D is replaced with its instrument Z.
- The 2SLS estimator is then

$$\hat{\delta}_{\text{IV,2SLS}} \equiv (\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'\mathbf{y}, \tag{9.31}$$

where $\mathbf{y}$ is an $n \times 1$ vector containing the outcomes y_i.

Graphs for IVs that identify LATEs

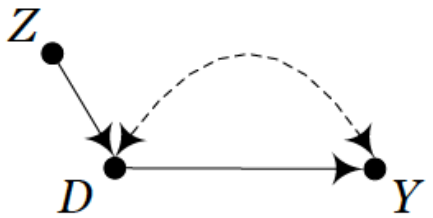
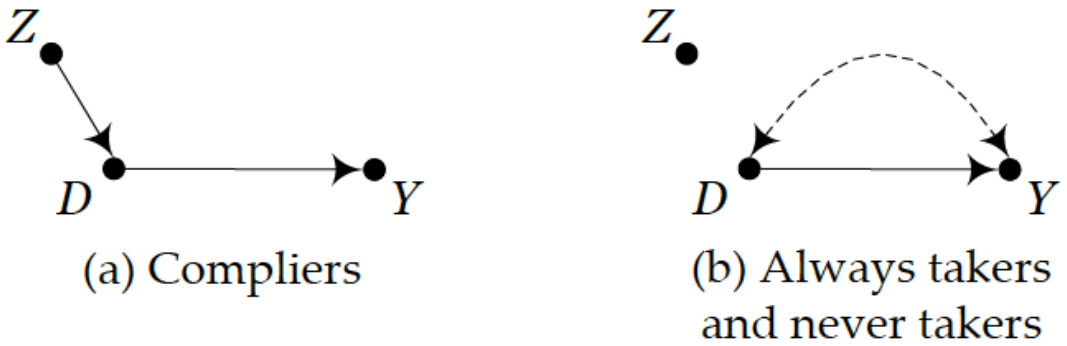


Figure 9.4 Instrumental variable identification of the causal effect of charter schools (D) on test scores (Y), where Z is the instrument.



(a) Compliers

(b) Always takers and never takers

Figure 9.5 Instrumental variable identification of the causal effect of charter schools (D) on test scores (Y), where separate graphs are drawn for compliers and noncompliers

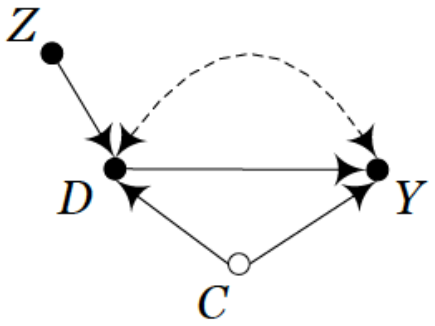


Figure 9.6 A combined graph for Figures 9.5(a)–(b), where Z is the instrument and compliance is represented as an unobserved latent class variable (C).

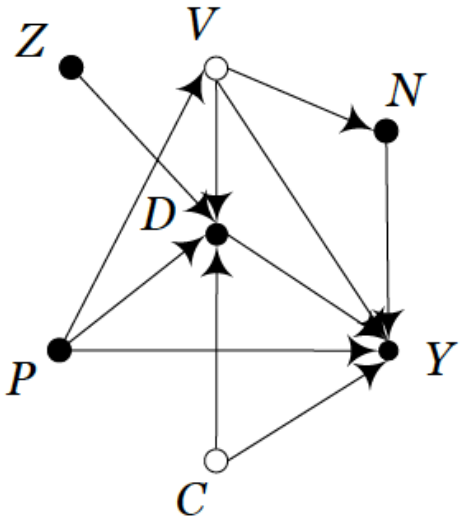


Figure 9.7 Identification of the LATE using an instrument (Z) for the charter school graph presented earlier in Figure 8.7. The unobserved variable V is a composite for the causal chain that generates self-selection in Figure 8.7 through information access and selection on the subjective evaluation of the individual-level causal effect.

Local IVs and Marginal Treatment Effects

- The marginal treatment effect (MTE) perspective can be easily grasped with only a slight modification to the setup of IV Demonstration 2.
- Instead of 10 percent of students receiving a voucher that is exactly equal to \$3,000, suppose instead that these 10 percent of students receive a voucher that is a random draw from a uniform distribution with a minimum of \$1 and a maximum equal to the tuition charged by the most expensive private school in the area.
- The size of each student's voucher is a valid instrument Z, maintaining the same assumptions as we did for IV Demonstration 2 (i.e., Z is randomly assigned, Z has a nonzero effect on D, and the effect of Z on D is monotonic).

- The monotonicity assumption is a little more complex than before, but it stipulates that the true probability of taking the treatment is higher for all individuals with values of Z equal to z'' rather than z' if $z'' > z'$. This fits cleanly into the notation introduced earlier by simply allowing Z to be many-valued.
- Heckman and his coauthors then define two related concepts: a local instrumental variable (LIV) and an marginal treatment effect (MTE).
  - An LIV is the limiting case of a component binary IV drawn from Z in which z'' approaches z' for any two values of Z such that $z'' > z'$.
  - Each LIV then defines an MTE, which is the limiting form of a LATE, in which the IV is an LIV.

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The impressive development of the IV literature in econometrics and statistics in the past two decades suggests a variety of recommendations for practice that differ from those found in the older IV literature:

- Weak instruments yield estimates that are especially susceptible to finite sample bias. Consequently, natural experiments should be avoided if the implied IVs only very weakly predict the causal variable of interest. No matter how seductive their claims to satisfy identification assumptions may be, resulting point estimates and standard errors may be very misleading.
- If individual-level causal effect heterogeneity is present, a potential IV should be used only if it satisfies a monotonicity condition. IV estimates should then be interpreted as LATE estimates.

- If individual-level causal effect heterogeneity is present, IVs should probably not be combined in a 2SLS model (except in the rare cases in which measures of all of the variables that account for the causal effect heterogeneity are available). Instead, IV estimates should be offered for those IVs that satisfy monotonicity conditions. These alternative estimates should then be interpreted as LATE estimates and reconciled with each other.
- When possible, IVs should be used to examine general patterns of causal effect heterogeneity. The possibilities for this type of analysis are most clearly developed in the literature on the identification of MTEs using LIVs.